Advanced Analytics Using Statistics – CCEE Cheatsheet

Sessions 11–23 Complete Study Guide

SESSION 11–12: SAMPLING AND ESTIMATION

Population vs Sample

- **Population**: Complete set of all units of interest
- **Sample**: Subset drawn from population to study
- **Parameter**: Numerical characteristic of population (μ, σ^2)
- **Statistic**: Numerical characteristic of sample ($\bar{x}$, s^2)

Univariate vs Bivariate Sampling

- **Univariate**: One variable per observation (e.g., salary only)
- **Bivariate**: Two variables per observation (e.g., age + income); allows study of relationships

Resampling and Bootstrap

- **Resampling**: Drawing repeatedly **from the sample** (not population)
- **Bootstrap**: Resampling **with replacement** to estimate variability of a statistic
- Purpose: Estimate standard error, confidence intervals when formulas complex or assumptions fail
- Key: Bootstrap sample size = original sample size; many iterations (1000s)

Sampling Error & Estimation

- **Point Estimate**: Single value to estimate parameter (e.g., $\bar{x}$ estimates μ)
- **Interval Estimate**: Range (confidence interval) where parameter likely lies
- **Sampling Error**: Difference between sample statistic and true population parameter

Central Limit Theorem (CLT)

Most important for MCQs:

For large sample size n (typically $n \geq 30$):

$$\bar{X} \sim \text{approx } N\left(\mu, \frac{\sigma^2}{n}\right)$$

- Mean of sample mean = μ (same as population)
- Standard deviation of sample mean = $\sigma/\sqrt{n}$ (standard error)
- Distribution of $\bar{x}$ approaches normal **regardless of original population shape**
- As n increases, distribution becomes more concentrated around μ

MCQ Alert: CLT is about **sampling distribution of sample mean**, not raw data

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SESSION 15–16: PREDICTIVE MODELLING & SUPERVISED SEGMENTATION

Informative Attributes

- **Informative**: Helps distinguish target classes or predict outcome
- Example: "Number of complaints" is informative for churn; "Customer ID" is not
- Used in feature selection before modelling

Supervised Segmentation

- Splitting data into segments using **input attributes** guided by **target label**
- "Supervised" = target guides splits
- Progressive: repeatedly split on best attribute for each subset

Decision Trees & Rules

- **Tree structure**: Root → internal nodes (splits) → leaves (predictions)
- **Split criterion**: Choose attribute that best separates classes (maximizes information gain or minimizes impurity)
- **Trees as rules**: Each root-to-leaf path = IF–THEN rule
    - Example: IF tenure < 6 months AND complaints > 3 THEN churn = "high risk"

Induction vs Prediction

- **Induction**: Learning rules/model from training data (building tree, finding splits)
- **Prediction**: Applying learned model to new unseen cases

Probability Estimation in Trees

At each leaf:

$$P(\text{class} = \text{k} \mid \text{leaf}) \approx \frac{\#\text{of class k in leaf}}{\text{total samples in leaf}}$$

- Estimated probability used for new cases falling in that leaf
- Often smoothed for robustness

Visualization of Segmentations

- **Tree diagram**: Shows attributes, splits, thresholds
- **Colored segments**: Scatter plots showing how splits separate classes
- Purpose: Interpret decision drivers and communicate to non-technical stakeholders

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SESSION 17: SIMULATION AND RISK ANALYSIS

Risk Analysis Basics

- **Risk analysis**: Identify uncertainties, quantify impact on outcomes, assess decision impact
- **Simulation**: Build model, play it out many times with random inputs to see outcome range

Monte Carlo Simulation

- Assign probability distributions to uncertain inputs (demand, price, cost, delays)
- Randomly sample these inputs many iterations (thousands)
- Compute outcome each time
- From distribution of outcomes: estimate probability of loss, worst-case, best-case, Value at Risk

Linear Optimization / Linear Programming

Key elements:

- **Decision variables**: Quantities to choose (x_1, x_2, ...)
- **Objective function**: Maximize or minimize (e.g., $Z = 5x_1 + 4x_2$)
- **Constraints**: Linear equalities/inequalities (e.g., $2x_1 + 3x_2 \leq 100$)
- **Feasible region**: Set of all solutions satisfying all constraints
- **Optimal solution**: Feasible solution giving highest/lowest objective value

MCQ Alert: Variables in LP objective are **decision variables**, not parameters or random variables

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SESSION 18–19: DECISION ANALYTICS & EVALUATION

Evaluating Classifiers

- **Accuracy** alone can mislead on imbalanced data (e.g., predicting all "non-fraud" may give high accuracy)
- **Precision** & **Recall** explicitly account for false positives and false negatives
- **F1 Score**: Harmonic mean of precision and recall
- **ROC–AUC / PR–AUC**: Area under curve metrics robust to class imbalance
- **Right metric depends on business**: fraud detection cares more about recall; spam filter cares about precision

Analytical Framework

Problem → Data & Features → Model → Evaluation → Business Impact

Baseline Model

- Simple reference model (e.g., "always predict majority class", current business rule)
- Used to check if complex ML actually adds value
- If fancy model only slightly beats baseline, complexity may not be justified

Diminishing Returns

- Better data/model complexity gives better metrics, but **not linearly**
- Beyond some point: big extra cost → tiny metric improvement → not worth it

Converting Metrics to Business Impact

- Map false positives / false negatives → rupees, risk, operational cost/benefit
- Decision: Is model performance improvement worth extra data/infrastructure investment?

SESSION 20–21: EVIDENCE AND PROBABILITIES – BAYES' RULE

Bayes' Rule Formula

$$P(H \mid E) = \frac{P(E \mid H) \cdot P(H)}{P(E)}$$

Key Terms

- **P(H)**: **Prior** – belief in hypothesis before seeing evidence
- **P(E | H)**: **Likelihood** – probability of evidence if hypothesis true
- **P(H | E)**: **Posterior** – updated belief after seeing evidence
- **P(E)**: Marginal likelihood or normalizing constant

Explicit Evidence Combination

With multiple pieces of evidence E_1, E_2, ...:

If **conditionally independent** given H (naive Bayes assumption):

$$P(E_1, E_2 \mid H) = P(E_1 \mid H) \cdot P(E_2 \mid H)$$

Each evidence provides push toward/away from hypothesis; combining = multiply pushes

Probabilistic Reasoning

- Express uncertainty as probabilities, not absolute yes/no
- Use conditional probabilities and Bayes' rule to connect hypotheses and evidence
- Output is probability or distribution, not deterministic statement

Common MCQ Traps

- **P(E | H) ≠ P(H | E)** (likelihood ≠ posterior) — be careful with order
- **Prior affects posterior**: Even strong evidence can't override very low prior
- **Naive Bayes assumes independence**: Real data may violate this

Numeric Bayes Example (Medical Test)

- Disease prevalence: P(D) = 0.02
- Test sensitivity: P(+ | D) = 0.95
- Test specificity: P(− | ¬D) = 0.90

If test is positive, what is P(D | +)?

$$P(D|+) = \frac{0.95 \cdot 0.02}{0.95 \cdot 0.02 + 0.05 \cdot 0.10} = \frac{0.019}{0.024} \approx 0.79$$

(Counterintuitive: Even with high test accuracy, many false positives when disease is rare)

SESSION 22: BUSINESS STRATEGY & COMPETITIVE ADVANTAGE

Competitive Advantage Definition

- Delivering **greater value** to customers **OR** same value at **lower cost** than competitors
- Sustainable = hard for others to copy quickly

Achieving Competitive Advantage via Analytics

Cost Leadership:

- Optimize operations, reduce waste, improve forecasting, streamline supply chains
- Analytics: demand prediction, inventory optimization, cost analysis
- Example: Walmart inventory data → right stock levels → lower costs

Differentiation:

- Personalize products, pricing, experiences so customers see higher value
- Analytics: recommendation systems, targeted offers, customer segmentation
- Example: Amazon recommendations, Netflix personalization

Sustaining Competitive Advantage

- Embedding analytics **deep in processes**: hiring, culture, IT systems (not just one tool)
- Continuous learning from new data, faster experiments than competitors
- **Ingredients**: Proprietary/hard-to-access data, strong infrastructure, talented team, strategic alignment

Investment in Data & Analytics

- Better data → better metrics → business KPIs (profit, risk, churn reduction)
- **Diminishing returns**: Extra data cost ≫ tiny metric gain = not justified
- Strategy question: Does analytics improvement justify investment cost vs competitor capabilities?

SESSION 23: FACTOR ANALYSIS & DIRECTIONAL ANALYTICS

Factor Analysis Basics

- **Goal**: Reduce many correlated variables into few underlying **latent factors**
- **Example**: Many exam scores → "quantitative ability", "verbal ability"

Key Concepts

- **Factor Loading**: Correlation between observed variable and factor; high loading = variable strongly reflects factor
- **Communality**: Proportion of a variable's variance explained by common factors
- **Eigenvalue**: Importance of a factor; often retain factors with eigenvalue > 1
- **Scree Plot**: Visual to choose number of factors

When to Use Factor Analysis

- Dimensionality reduction before further modelling
- Identifying latent constructs in multivariate data
- Summarizing many correlated variables

Directional (Circular) Statistics

- **Data type**: Directions (angles 0°–360°) on circle, or points on sphere
- **Examples**: Wind direction, animal movement, time of day (as angle)

Key Issue with Directional Data

- **0° and 360° are the same direction**
- Ordinary arithmetic mean fails: mean of (1°, 0°, 359°) ≠ 120°; it should be near 0°
- Need special **circular mean** that wraps around

Directional Analytics Concepts

- **Mean direction**: Circular analogue of mean, accounting for wrap-around
- **Concentration**: How tightly directions cluster (like standard deviation for circles)
- **Circular distributions**: von Mises distribution (analog of normal for circular data)

KEY MCQ TIPS FOR CCEE

1. **CLT is about sampling distribution of sample mean**, not raw data
2. **P(E | H) ≠ P(H | E)** — know the order; likelihood ≠ posterior
3. **Baseline model**: Simple reference to judge if complex model is worth complexity
4. **Imbalanced data**: Accuracy misleading; use precision, recall, F1, AUC
5. **Factor loading**: **Correlation** between variable and factor (not variance)
6. **Directional data**: Must use circular statistics (0° = 360°)
7. **Bootstrap**: Resampling **with replacement** from observed sample
8. **Supervised segmentation**: **Target guides splits** (supervised), not unsupervised
9. **Bayes on imbalanced priors**: Even strong evidence can't overcome very low prior
10. **Diminishing returns**: Big cost for tiny metric gain often not justified

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QUICK FORMULA REFERENCE

Concept	Formula	Notes
Standard Error	$SE = \sigma/\sqrt{n}$	How much sample mean varies
CLT (large n)	$\bar{x} \sim N(\mu, \sigma^2/n)$	Asymptotic normality
Bayes' Rule	$P(H \mid E) = P(E \mid H) \cdot P(H) / P(E)$	Order matters: $E \mid H \neq H \mid E$
Communality	Proportion of var explained by factors	0 to 1 range
Factor Loading	Correlation(variable, factor)	High = strong reflection
Sampling Error	Sample Statistic − Parameter	Expected deviation
Bootstrap SE	Std dev of bootstrap statistic	Distribution of estimates

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QUICK DECISION TREE

Is data directional (0°–360° angles)? → YES = use circular statistics
Do you have many correlated variables? → YES = consider factor analysis
Are classes imbalanced (10% positive)? → YES = use precision/recall/F1, not just accuracy
Is Bayes' prior very low? → YES = evidence won't overcome it
Did fancy model barely beat baseline? → YES = probably not worth extra cost
Are you sampling for statistics? → YES = watch out for sampling error and CLT conditions

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TYPICAL MCQ PATTERNS

Pattern	Answer Clue
"Population is …"	Complete set of all units; "parameter" not "statistic"
"Univariate vs bivariate"	Univariate = 1 variable; bivariate = 2 variables
"Bootstrap resampling …"	Drawing **from the sample** with **replacement**
"Factor loading …"	**Correlation** between variable and factor
"Bayes' rule helps update …"	**Posterior probability** after observing evidence
"Directional data …"	0° = 360°; use **circular mean**, not arithmetic mean
"Baseline model …"	Simple reference to judge if complex model adds value
"Optimize cost under constraints"	Linear programming problem
"Reduce many variables to few"	Factor analysis or PCA
"Competitive advantage sustained by"	Proprietary data, strong infrastructure, continuous learning

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REVISION ORDER FOR CCEE

1. **Start with Sessions 11–12** (Sampling, CLT) — foundational
2. **Then Sessions 15–16** (Trees, segmentation) — predictive modelling
3. **Sessions 17** (Simulation, LP) — decision-making tools
4. **Sessions 18–19** (Evaluation, baseline) — how to judge models
5. **Sessions 20–21** (Bayes) — probabilistic reasoning
6. **Session 22** (Business strategy) — linking analytics to impact
7. **Session 23** (Factor analysis, directional) — specialized techniques

Practice MCQs after each session, then do mixed sets.

Compiled for CCEE Advanced Analytics using Statistics exam. Keep this handy during revision!